

4	(a) work done moving unit positive charge from infinity to the point	M1	A1	[2]
	(b) (i) $x = 18 \text{ cm}$	A1	[1]	
	(ii) $V_A + V_B = 0$ $(3.6 \times 10^{-9}) / (4\pi\epsilon_0 \times 18 \times 10^{-2}) + q / (4\pi\epsilon_0 \times 12 \times 10^{-2}) = 0$ $q = -2.4 \times 10^{-9} \text{ C}$ <i>(use of $V_A = V_B$ giving $2.4 \times 10^{-9} \text{ C}$ scores one mark)</i>	C1	C1	[3]
	(c) field strength = (–) gradient of graph force = charge $\times$ gradient / field strength <i>or</i> $\text{force} \propto \text{gradient}$ force largest at $x = 27 \text{ cm}$	B1	B1	[3]
5	(a) at $t = 1.0 \text{ s}$ $V = 2.5 \text{ V}$	C1		